Module 4 Basic Content:

Why humans need water

- Homeostasis: the process of maintaining internal balance among all the different systems in an organism; water is at the heart of homeostasis
- On average, most humans can live only 3 days without water before risking death
- On average, the human body is composed of about 55% water
- Organs and tissues:
    - Lungs, muscles, and kidneys – 80% water
    - Brain and heart – 70% water
    - Skin – 65% water
    - Blood – 50% water
    - Bones – 30% water
- Water is found in and around all human cells and is required for proper functioning.
- Water is the base of almost all liquids produced by the human body (e.g., saliva, mucus, blood, urine, sweat, tears)
    - Odd factoid: Two main systems (respiratory and digestive) in the human body are lined with mucous membranes (meaning they produce mucus – think snot or phlegm). Mucus serves many different purposes. In the digestive system it ensures food can be chewed and swallowed properly, begins digestion, moves food boluses smoothly into the stomach, and provides a protective layer in the stomach and small intestine to prevent stomach acid from damaging them. In the respiratory system (sinuses and lungs) mucus keeps the lungs moist so the cells in the alveoli can exchange oxygen and carbon dioxide, as well as being the first line of immune defense against inhaled germs which get stuck in the mucus and are removed through coughing and blowing the nose. None of these processes function properly when the body is dehydrated.

        - Dehydration is the state of having less water in your body than is needed for normal functioning; when dehydrated, humans feel dizzy, tired, have muscle aches and headaches, and struggle to think clearly.
            - Persistent dehydration often results in kidney stones.
            - Extreme dehydration produces convulsions, loss of consciousness, and can lead to death. Low blood volume 1) increases blood acidity which damages blood vessels, 2) decreases kidney function because there's not enough water to take away toxins filtered out by the kidneys, and 3) produces severe electrolyte imbalance which shuts down normal cellular processes (oxygen/carbon dioxide exchange, protein production) leading to cell death, heart attacks and organ failure.
    - To maintain homeostasis, anyone 9 years and older needs to consume 2-3 liters of water per day. Those engaged in physical activities need more than this to replenish water lost through increased respiration and sweat.

Historical reasons drinking water laws were enacted

Drinking water laws

- Safe Drinking Water Act: SDWA is a law designed to protect the water we drink. The law was originally passed by Congress in 1974 and has been updated twice
- National Primary Drinking Water Regulations:
- National Secondary Drinking Water Regulations:
- Unregulated Contaminants:
- Bottled Water Regulations: